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CHARACTERISTICS OF NERVE FIBRES

INTRODUCTION TO NERVE CELLS



Neuron (nerve fibres)



Glial cells

Neuron : Functional units of central nervous system.

Glial cells : Supporting cells of nervous system (Numerous).

Neurons : Glial cells :: 1 : 10.

Glial Cells

00:02:06

CLASSIFICATION :

Based on size

- Macroglial cells
  - Astrocytes
  - Oligodendrocytes
  - Schwann Cells
  - Ependymal cells
  - Satellite cells
- Microglial cells
  - Microglia

MICROGLIA CELLS :

Function : Modified immune cells.

Precursor : Monocyte macrophage system.

ASTROCYTES :

Star shaped cells.



Astrocytes



Anti-GFAP antibodies

Types :

- Protoplasmic : In grey matter.
- Fibrous : In white matter.

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Significance :

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GFAP (Glial fibrillary acidic protein) :

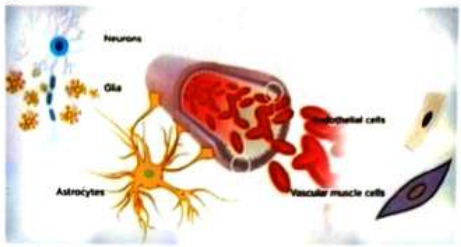
- It is an intermediate filament.
- Anti GFAP antibodies : Added to visualize astrocytes (Tumour marker).

Functions :

1. Blood Brain Barrier (BBB) : most critical.

Structures involved :

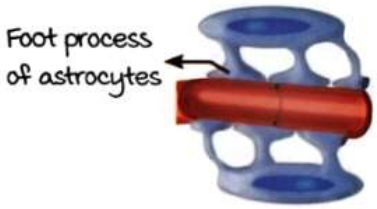
- Endothelial cells : Tight junctions.
- Astrocytes : Foot process.



Blood brain Barrier

2. Migration of neurons :

Radial glial cells (Astrocytes) : Forms pathway for neuronal migration during development.



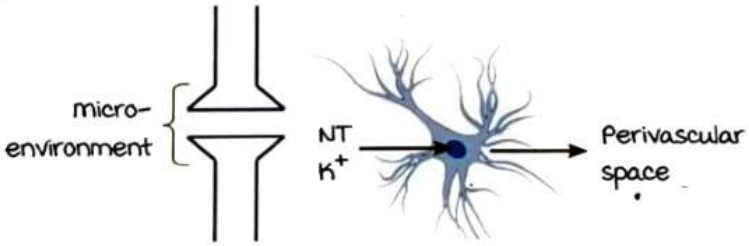
BBB

3. Synapse formation :

Due to growth factor : Thrombospondin (Released from astrocytes).

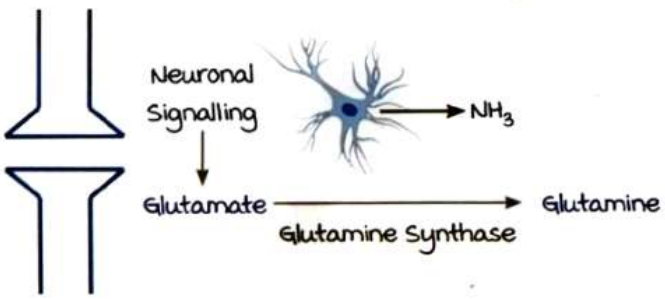
4. Environmental modulation (Synaptic cleaning) :

Neurotransmitters and ions from neuronal microenvironment → Cleaned by astrocytes → Into perivascular space.



Synaptic clearing

5. Detoxification of ammonia (Glutamate – Glutamine cycle) :



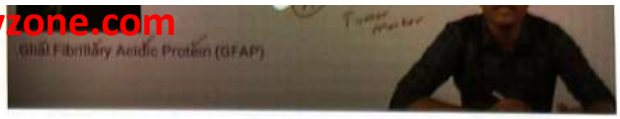
Note : ↑NH_3 is a toxic substance to the brain.

6. Neurovascular coupling :

- Principle of Functional MRI : Functional hyperemia (↑ blood flow to active areas of the brain).

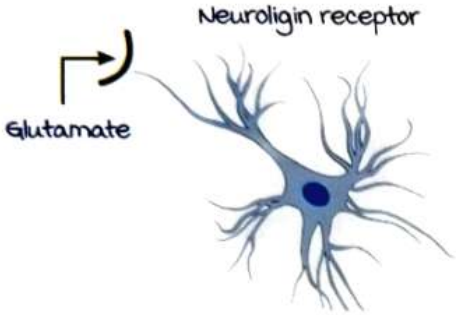
Feedback

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- Glutamate → Act on neuroligin receptors of astrocytes → ↑Vasodilators (Arachidonic acid derivatives, prostaglandins) → ↑blood flow.



→ Arachidonic acid derivative Prostaglandins →



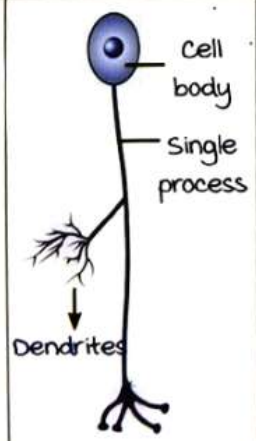
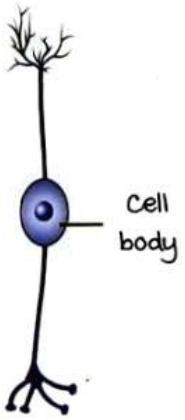
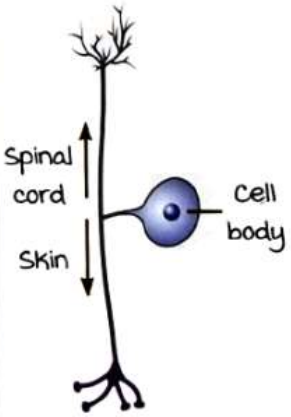
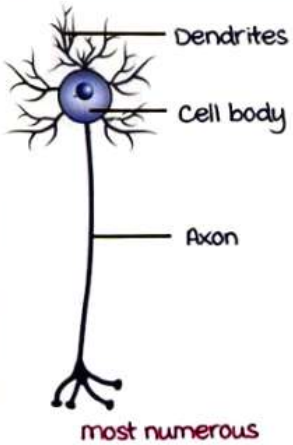
Functional MRI

OLIGODENDROCYTES AND SCHWANN CELLS :

	Oligodendrocytes	Schwann Cells
myelinate	Multiple axons	Single axon
Location of neurons	CNS	PNS

Neurons 00:16:30

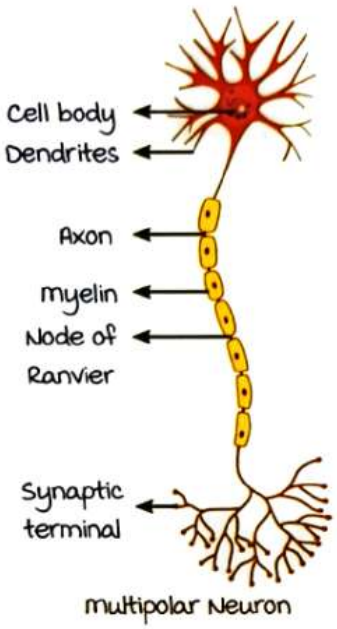
CLASSIFICATION :

	Unipolar	Bipolar	Pseudounipolar	Multipolar
Number of processes from cell body	Single (Dendrite/axon)	2	Single process split into 2	Multiple
Location of neuron	Invertebrates	Retina	Dorsal root ganglia of spinal cord	Spinal ∝ motor neuron
Image				 most numerous

Feedback

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MULTIPOLAR NEURON



multipolar Neuron

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Structure :

1. Cell body (Soma) :

Contains 2 structures :

- Nucleus : Controls functions of neuron.
- Nissl bodies (Rough endoplasmic reticulum) : Synthesize proteins.

Affected during nerve cell injury.

2. Dendrites :

- Finger-like projections.
- Receiving end of stimulus in neuron.

3. Axon :

- Axon hillock : Part of cell body giving rise to axon.
- Initial segment : First proper beginning of axon.
- Conduction of electrical signals : Based on myelin
  - Fast : Myelinated.
  - Slow : Unmyelinated.
- Node of Ranvier :
  - Part of axon devoid of myelin.
  - Maximum Na^+ channels present (2000-12000/sq.m) : For action potential.

4. Synaptic terminal (Axonal terminal) :

Ending of a neuron.

Significance :

Most important electrical activity : Action potential.

Site of initiation of action potential :

- Motor neuron : Axon hillock + Initial segment.
- Sensory neuron : 1st Node of Ranvier.

Factors affecting Nerve Conduction Velocity (NCV) :

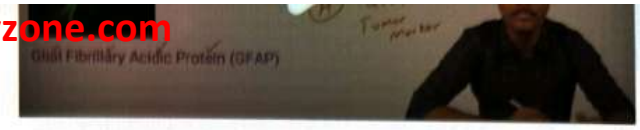
1. Myelin :

- Provides insulation : Thickness of neuron ∝ Nerve conduction velocity.
- Composition :
  - Lipids (80%) : Sphingomyelin.
  - Proteins (20%) : Myelin Basic Protein (MBP), zero protein.
- Clinical significance :

☐ Multiple Sclerosis (Demyelinating disorder) : Autoantibodies against MBP → Break down myelin → ↓NCV.

Feedback

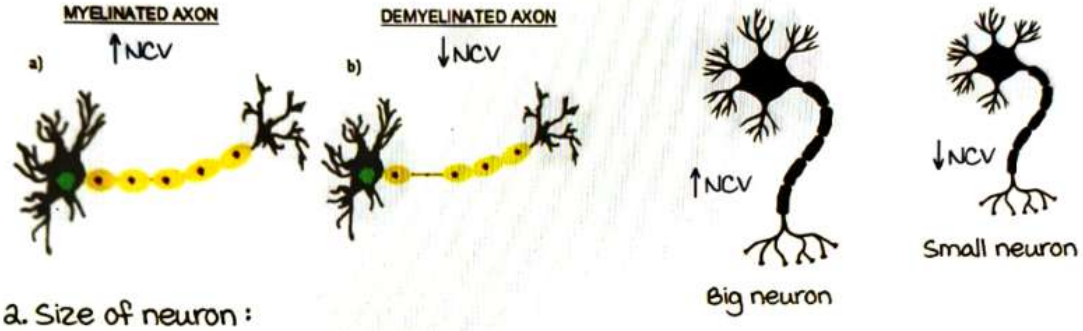
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----- Active space -----

myelinated axon :

- ↑NCV.
- ↑Axonal resistance : Ions move only through node of ranvier (↓loss).
- ↓membrane capacitance : Ability to store charges ↓ → Ions used for action potential. $\left(\text{Thickness} \propto \frac{1}{\text{Capacitance}}\right)$



2. Size of neuron :

Diameter ∝ nerve conduction velocity.

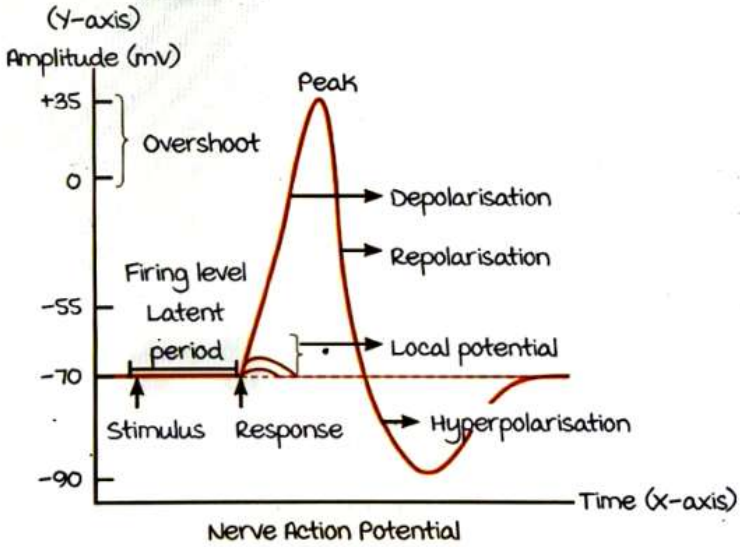
3. Temperature :

Temperature ∝ nerve conduction velocity.

Action Potential of Neuron

00:35:37

PHASES



Nerve Action Potential

Latent Period :

It is the time gap between stimulus and response.

	Phases	Change in Potential	Ion changes
I	Local potential	-70mV (RMP) → -55mV (Firing level)	Slow Na^+ influx (Less)
II	Depolarization	-55mV (Threshold) → +35mV	Fast Na^+ influx (more) : Opening of voltage gated Na^+ channels (↑intracellular positive charge)
	Overshoot of action potential	0mV → +35mV	
III	Repolarization	+35mV → -70mV	K^+ efflux
IV	Hyperpolarization	-70mV → -90mV	• Slow closure of K+ channel. • Influx of Cl^- ions.

Feedback

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Note :

E_{Na} = +60mV but during action potential, potential reached is ≈35mV.

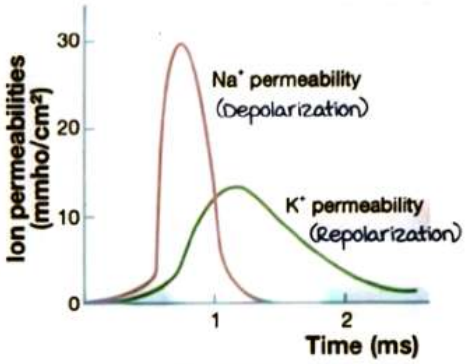
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Clinical significance of hyperpolarization :

Principle : Hyperpolarization (↑intracellular -ve charge) → Inhibits neuronal function.

Physiology : GABA → Inhibitory neurotransmitters (D/t neuronal hyperpolarization).

Applied anatomy : GABA agonists (For sedation).
- Barbiturates.
- Benzodiazepines.



Change in ion permeability

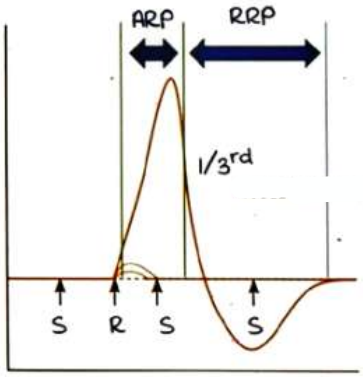
REFRACTORY PERIOD :

1. Absolute Refractory Period (ARP) :

- From threshold point to 1/3rd repolarisation.
- On second stimulus → No response in neuron.
- Inactivated voltage gated Na^+ channels.

2. Relative Refractory Period (RRP) :

- From 1/3rd repolarization to end of action potential.
- On second stimulus of maximum strength → Response seen.

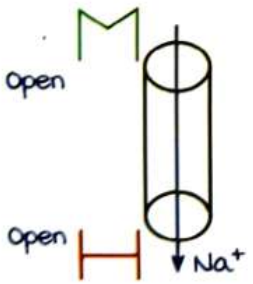


Refractory periods

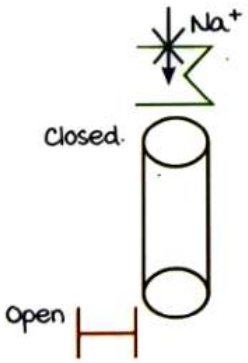
Note :

Voltage gated sodium channels :

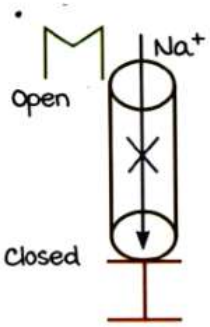
- m gate : Activation gate.
- H gate : Inactivation gate.



Open state



Closed state

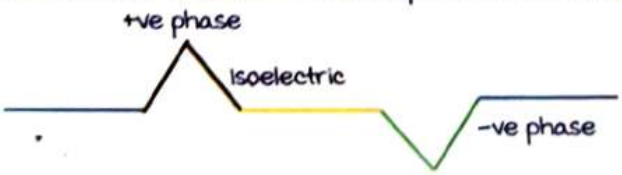


Inactivation state

TYPES OF ACTION POTENTIAL :

Biphasic action potential :

Electrode placed on surface of neuron → Action potential recorded.



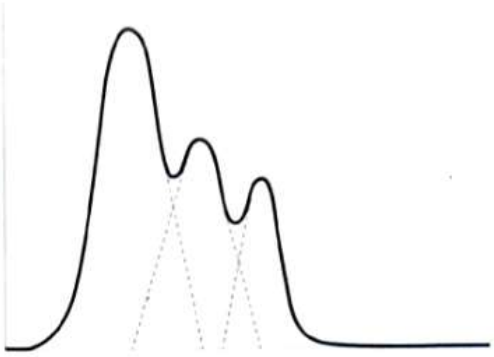
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Compound action potential :

- multiple peaks.
- In mixed nerve : D/t many nerve fibres & varying conduction velocity.



Compound action potential

Graded potential (GP) vs Action potential (AP) :

	Graded potential	Action Potential
Cause	Local potential	Summation of local potential → Reach threshold → AP
Summation	Possible	Not possible
Threshold	Not needed	Needed
Amplitude of potential	Changes	Constant
All or none Law	Does not obey	Obeys

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CLASSIFICATION OF NERVE FIBRES

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Classification of nerve fibers

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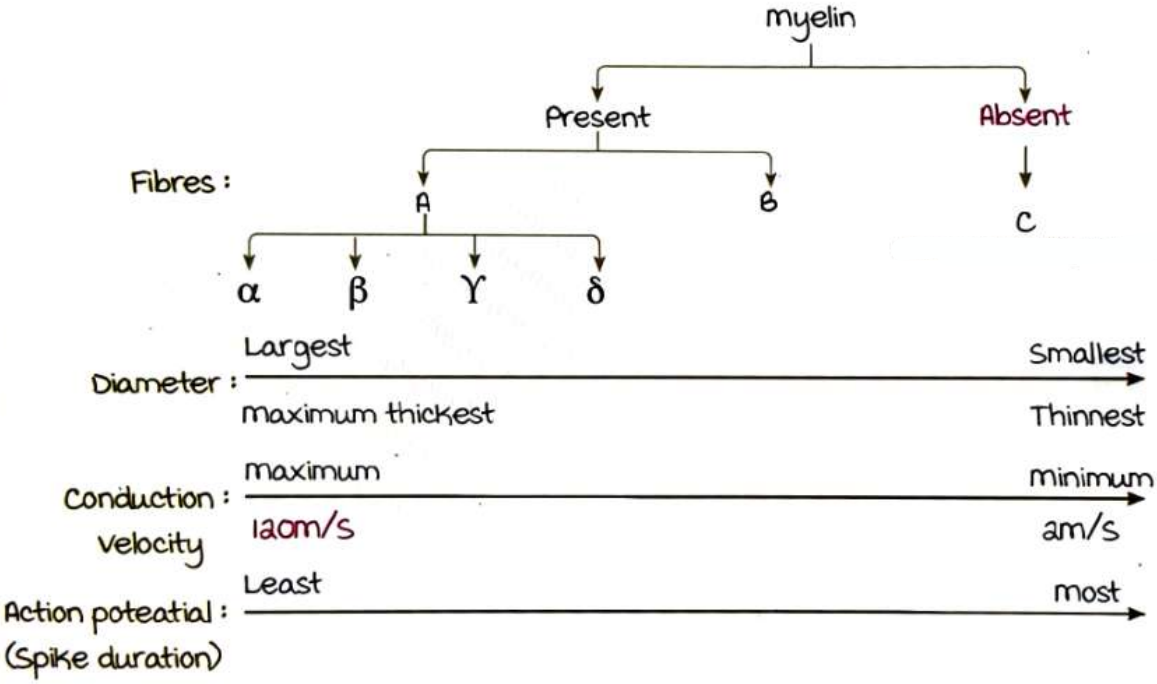
Types :

1. Erlanger-Gasser classification.
2. Lloyd-Hunt classification.

Erlanger Gasser Classification :

Erlanger and Gasser received Nobel Prize.

Classified based on myelin.



Functions of nerve fibers :

Fibers are sensory (S), motor (M), or both.

Type	Function
Aα	S : Proprioception (Joint position sense)
	M : Alpha motor neuron (Spinal cord)
Aβ	S : Touch sensation
Aγ	M : muscle spindle (Gamma motor neuron in spinal cord : Regulate tone)
Aδ	S : Fast pain (1st pain)
B	Preganglionic autonomic nerve fibers
C	S : Slow pain (2nd pain)
	Postganglionic autonomic nerve fibers

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Note :

- Susceptible to compression : Aα (Largest) > Aβ.
- Susceptible to hypoxia :
  - most : B.
  - Least : C.

LLOYD-HUNT CLASSIFICATION

AKA sensory classification : Only for sensory fibres.

Numerical classification :

Type		Corresponding to Erlanger-Gasser classification
I	Ia : Muscle spindles	Aα
	Ib : Golgi Tendon organ	
II		Aβ
III		Aδ
IV		C

Action Potential 00:17:35

CHARACTERISTICS :

For generating response : Stimulus of particular { Strength. / Duration.

STRENGTH-DURATION CURVE :

Rheobase :

Minimum strength of current required to produce response.

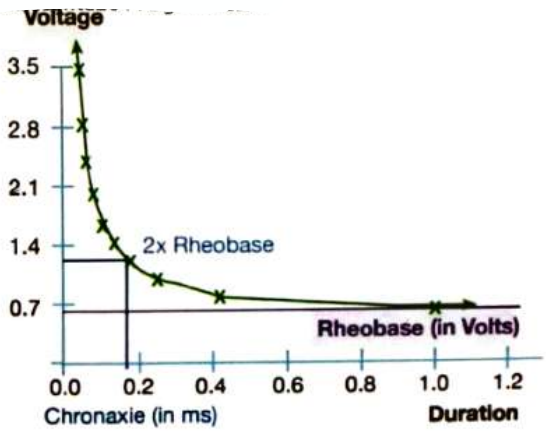
Chronaxie :

Minimum time required to produce response with twice rheobase strength.

Importance :

Chronaxie α $\frac{1}{\text{Excitability}}$

	Nerve	Muscle
Chronaxie	↓	↑
Excitability	↑	↓



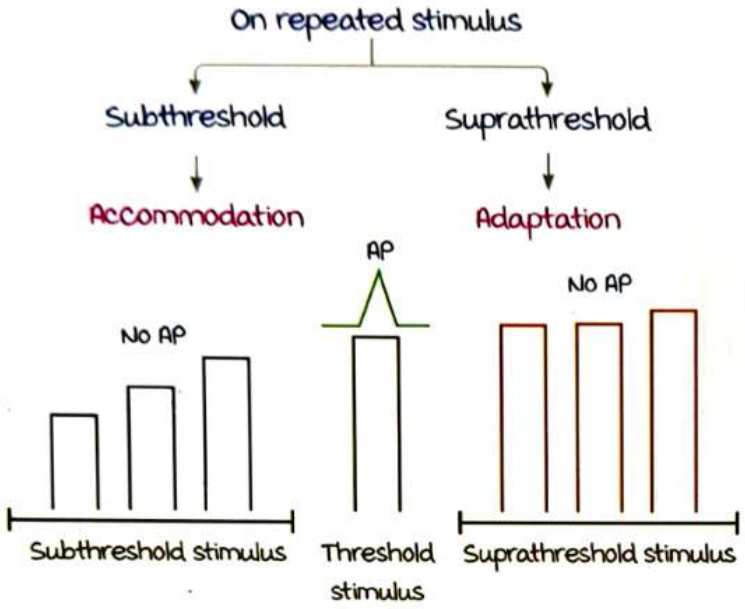
Strength - duration graph

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RESPONSE OF ACTION POTENTIAL :

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Nerve Injury

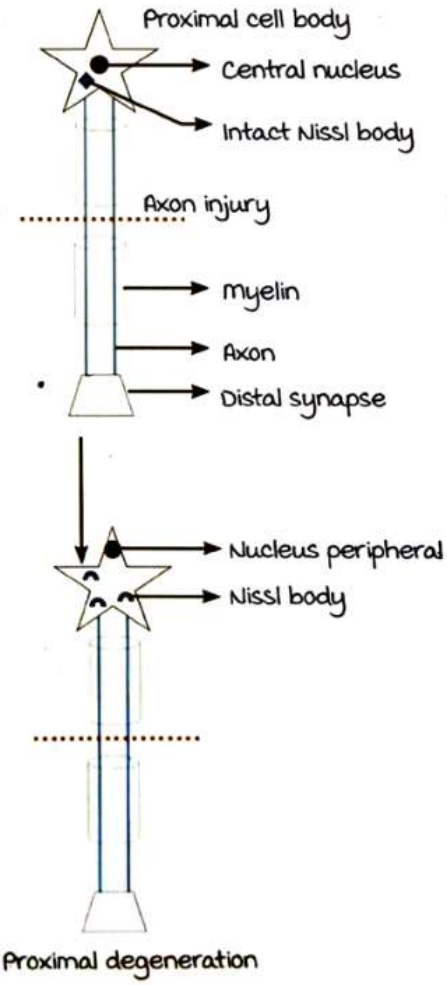
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m/c cause : Bone fractures.

m/c part susceptible to injury : Axon.

Based on the nature of injury :

- Degeneration.
- Regeneration.



DEGENERATION :

Distal Degeneration :

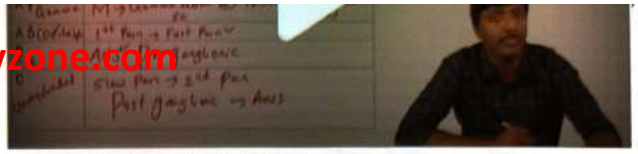
- AKA Wallerian Degeneration.
- Earliest (within 24 hours of injury).
- Changes :
  a) Axonal degeneration.
  b) myelin degeneration.
  c) Phagocytosis of debris : Clearing site of injury.

Proximal Degeneration :

- 36 hours after injury.
- Changes :
  - Nucleus : Pushed to periphery.
  - Nissl body : Chromatolysis (Broken down).

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REGENERATION :

- 90 hours after injury.
- Slow process : 1 - 3mm/day.
- Tinel sign :
  - Procedure : Tapping at site of injured nerve.
  - Inference : Tingling sensation (Electric shock type) felt → Indicate good prognosis.

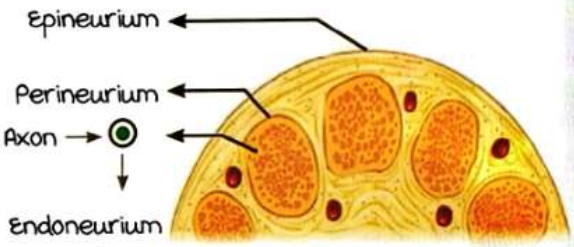


Tinel sign

Note :

Encapsulating structures :

- Epineurium : Outermost layer.
- Perineurium : Encloses bundles, axon.
- Endoneurium : Encloses axons.



Coverings of neuron

CLASSIFICATION OF NERVE INJURY :

To clinically assess prognosis and plan treatment.

1) Seddon's classification :

	Neuropraxia	Axonotmesis	Neurotmesis
Features of injury	• D/t compression. • Intact nerve. • Local conduction block (+). • Eg : Saturday night palsy.	• Axonal injury. • Wallerian degeneration. • Regeneration : 1-3mm/day.	Complete transection
Recovery	Spontaneous	Possible	-
Prognosis	Good	Intermediate	Worst

2) Sunderland classification :

Grade	Types of injury
I	Neuropraxia
II	Axonotmesis
III	Axonotmesis + Endoneurium disrupted
IV	Axonotmesis + endoneurium and perineurium disrupted
V	Neurotmesis (Axon + Epi + Peri + Endo disrupted) : Worst prognosis

In Grade IV & V : Surgical intervention.

Feedback